Two-stage Stewart platform: ManifoldsX and MuJoCo against forward kinematics

manifoldsx.github.io/stewart-benchmark

A two-stage Stewart platform is driven by its 12 leg lengths for 18 s. With every joint closed, each platform's pose is pure geometry, so the exact answer is known without running any dynamics. Over the full run, the worst top-platform error of ManifoldsX is 0.0000008 mm at 13–14× real time on one core. MuJoCo, with the stiffest constraint settings it accepts, reaches 0.11 mm at 1 ms steps (16.6×) and 0.0031 mm at 0.25 ms (4×); with its default settings the error is 43 mm at any step size. One model and one load case; the limits are listed in section 5.

1. Results

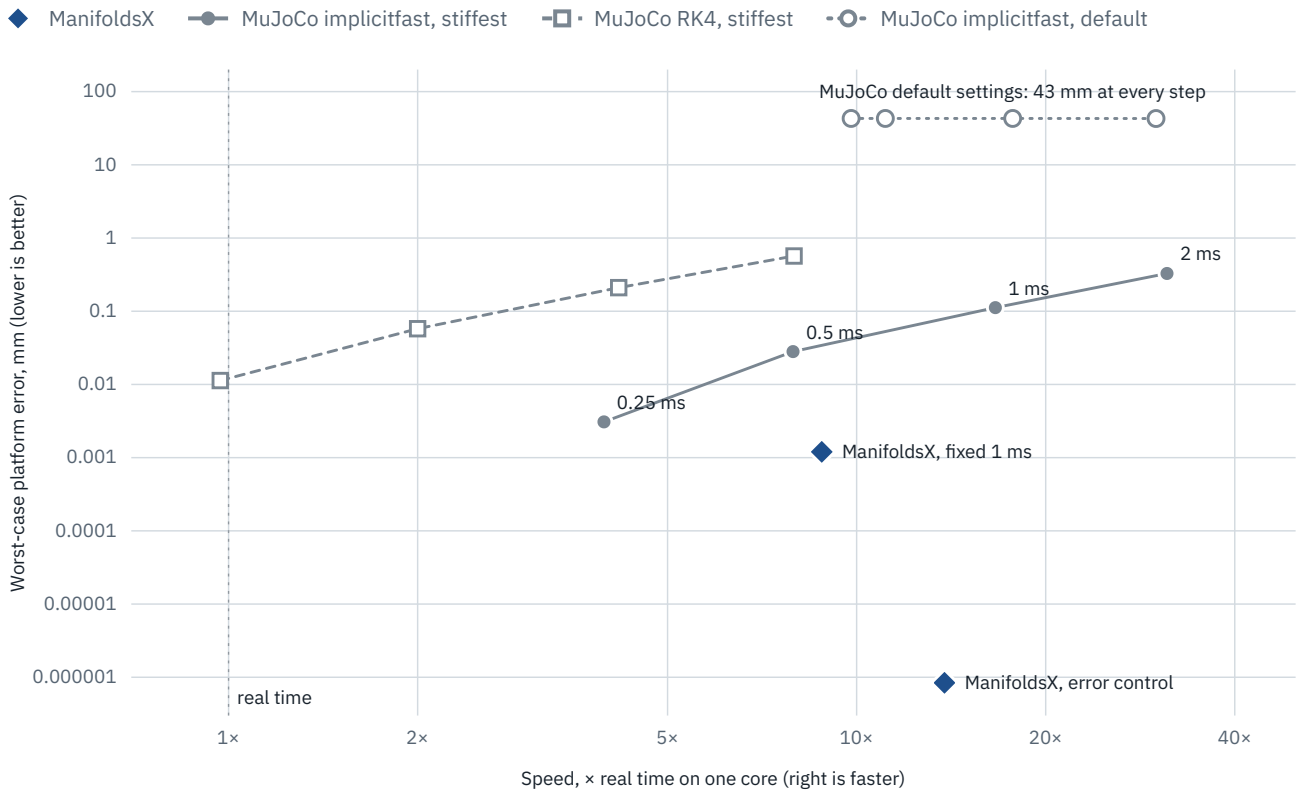


Figure 1. Speed against worst-case top-platform error. Both axes are logarithmic; right is faster, lower is more accurate. Labels on the MuJoCo points are step sizes.

Table 1. Worst values over the 18 s run, single core

Setup	Speed (× real time)	Loop gap (mm)	Leg-length error (mm)	Platform error (mm)
ManifoldsX, HHT index-3, error tolerance 1e-5, steps up to 3 ms	13–14	0.00000001	exact	0.0000008
ManifoldsX, fixed 1 ms steps, Newton tolerance 1e-6	8–9	0.0008	exact	0.0012
MuJoCo implicitfast, 2 ms, stiffest	31	0.006	0.20	0.33
MuJoCo implicitfast, 1 ms, stiffest	16.6	0.0017	0.069	0.11
MuJoCo implicitfast, 0.5 ms, stiffest	7.9	0.0004	0.017	0.028
MuJoCo implicitfast, 0.25 ms, stiffest	4.0	0.0001	0.0016	0.0031
MuJoCo RK4, 1 ms / 0.25 ms, stiffest	4.2 / 1.0	0.0008 / 0.00006	0.13 / 0.007	0.21 / 0.011
MuJoCo implicitfast, 1 ms, default settings	17.7	6.0	7.1	43

"Exact": ManifoldsX drives the leg lengths as constraints, so they hold to its Newton tolerance. Stiffest MuJoCo settings: constraint time constant = 2 × step (MuJoCo's lower limit) and solimp 0.9999. Default: 20 ms and 0.9 / 0.95.

2. Method

Model Two stacked Stewart platforms: 26 moving bodies, 24 spherical and 12 prismatic joints. The 12 leg lengths follow tables (spline-smoothed, sampled every 1 ms, linear between samples) for 18 s. Both solvers read the same model file.

ManifoldsX Index-3 HHT ($\alpha = 0.3$) with a full Newton matrix and local error control at 1e-5, steps up to 3 ms. Joints and leg drives are exact constraints.

MuJoCo 3.15 Same bodies, masses, inertias and joint points. Each loop closes with an equality connect. Each leg follows its table through a joint equality coupled to a virtual clock joint, so the constraint sees the target's velocity as well as its position. A plain C loop around mj_step refreshes the targets every step.

Reference Forward kinematics. Both stages are solved from the 12 leg lengths with Newton's method at every saved time, to 1e-14 m, with no dynamics and no solver involved.

Metrics Worst case over the run of the top platform's position error against the reference, the largest loop gap, and the largest leg-length error.

Cost Single-core wall time for the 18 s run on one desktop machine.

3. Why the errors differ

ManifoldsX solves joint and drive positions at every step, so its error is set by its Newton tolerance and does not grow with the step size. MuJoCo enforces constraints at acceleration level with stabilization. Each constraint trails by roughly acceleration × time constant^2, and the time constant cannot go below twice the step, so its error falls with the square of the step.

4. Second computer

A second computer gave the same errors and similar speed ratios.

5. What this result does not show

- **One model, prescribed motion.** It measures closed-loop kinematic accuracy under driven motion. It says nothing about contact, force-driven dynamics, or throughput across many parallel environments, which is what MuJoCo is designed for.
- **MuJoCo has no built-in time-driven joint constraint.** The clock-joint drive is built from standard MuJoCo features. A better MuJoCo setup for driven loops is welcome and will be added.
- **ManifoldsX itself is not public.** The models, the MuJoCo scripts, the forward-kinematics reference and the saved trajectories are available on request.
- **One machine for the published speeds.** See section 4.

6. Corrections

An earlier internal version of this benchmark put ManifoldsX at 0.0004 mm. That figure came from scoring against ManifoldsX's own 0.25 ms run through a cubic spline, which overshoots at the 1 ms kinks of the piecewise-linear leg tables. Scored against forward kinematics, it was 0.00002 mm.

Later, ManifoldsX began accepting a step only when every constraint meets its tolerance; before, only the average over all constraints had to. Same model and settings, re-run: the error-controlled run went from 0.00002 to 0.0000008 mm and the fixed 1 ms run from 0.0097 to 0.0012 mm, at the same speed. The MuJoCo numbers did not change in either case.

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